

GED Math Formula Sheet

Perimeter

square: $P = 4s$	rectangle: $P = 2L + 2W$	triangle: $P = s_1 + s_2 + s_3$
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Area

square: $A = s^2$	rectangle: $A = LW$	triangle: $A = \frac{1}{2}bh$
circle: $A = \pi r^2$	parallelogram: $A = bh$	trapezoid: $A = \left(\frac{1}{2}\right)h(b_1 + b_2)$

Surface Area and Volume

rectangular/right prism	$SA = ph + 2B$	$V = Bh$
cylinder	$SA = 2\pi h + 2\pi r^2$	$V = \pi r^2 h$
pyramid	$SA = \left(\frac{1}{2}\right)ps + B$	$V = \left(\frac{1}{3}\right)Bh$
cone	$SA = \pi rs + \pi r^2$	$V = \left(\frac{1}{3}\right)\pi r^2 h$
sphere	$SA = 4\pi r^2$	$V = \left(\frac{4}{3}\right)\pi r^3$

Algebra

slope of a line	$m = \frac{(y_2 - y_1)}{(x_2 - x_1)}$
equation of a line (slope-intercept)	$y = mx + b$
equation of a line (point-slope)	$y - y_1 = m(x - x_1)$
quadratic equation (standard form)	$y = ax^2 + bx + c$
quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
pythagorean theorem	$x = a^2 + b^2 = c^2$
simple interest	$I = prt$
distance formula	$d = rt$
total cost	number of units $\times$ price per unit

(I = interest, p = principal, r = rate, t = time)

Data

Mean	The total of the values of a data set, divided by the number of elements in the data set
Median	The middle value in an odd number of ordered values of a data set or the mean of the two middle values in an even number of ordered values in a data set

*Note: b = length of base; B = area of base